TEAM 18

SPL - I (SWE 4303)

LabTrack

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Project overview :

– A Research Lab Management System

A management system that organizes experiments, samples, inventory, and equipment reservations in a research lab

Tracks experiments with git like **version control**

Handles **samples**, **inventory**, **equipment** & room **booking**

Prevents **mistakes** and missing **records**

Keeps the lab **organized** & **efficient**

KEY Features :

- **Secure User Management :**

Multiple User can **Login** to their account & do their **ROLE BASED** permitted work.

[Admin]
|

| | |
Researcher **Manager** **Technician**
Logs Experiment | Approves Protocols | Updates Inventory

ADMIN can :
Create, View, Promote, Delete Users.

RESEARCHER can :
Add, View, Modify, Delete experiments, Book Labs

MANAGER can :
Approve Reservation, View Report

TECHNICIAN can :
Add, View, Update Inventory

KEY Features :

- **Version Control :**

  keeping **track** of every **changes** made in experiment or protocols(inspired by **git**)

  - **CREATE** new version while editing
  - **VIEW** old versions
  - option for **RESTORING** edits
  - maintain **HISTORY** of changes

- **Sample Tracking :**

  Track **where** each sample came from, how it is **used** or **modified**

- **Protocol Feature :**

  Stores all the Standard Operating Procedures (**SOP**), **Lab Manuals**, **User Guide** & can be **edited** or **updated**.

- **Protocol Feature :**

  Track **real-time quantity** of each reagent.

- **Protocol Feature :**

  Check Lab Room availability & reserve it at specific dates and time slots for experiments, meetings, or group work.

Previously Proposed timeline :

Group Form, Supervisor Select, Draft Proposal

IMPLEMENTATIONS

Auth, File I/O, Utilities

Version Control, Protocol, Inventory, Equipment, Report Service, Lab Booking, Samples

Activity, Weeks	2	3	4	5	6	7	8	9	10	11	12	13	14
Group	■	■											
Proposal			■	■									
Design				■									
Framework					■								
Modules						■	■						
Integration								■					
Development									■	■	■	■	
Final													■
Testing						■	■	■	■	■	■	■	■

Our Progress :

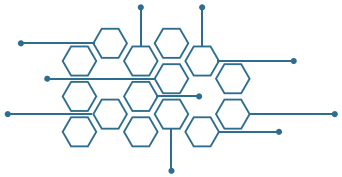
Completed Modules

- **Users :** Users Can Login, Logout, Do their ROLE-BASED works
- **AuthService :** Unique User Authetification.
- **Experiment :** Add, Edit, View experiments.
- **Inventory :** Show quantity of items, Add, Edit items.
- **Equipment :** Used equipments in Lab, Reserve for experiments.
- **Lab Booking :** Books Lab Rooms distinctly.
- **Report Service :** Gives the Summary of experiments.

Testing & Documentation :

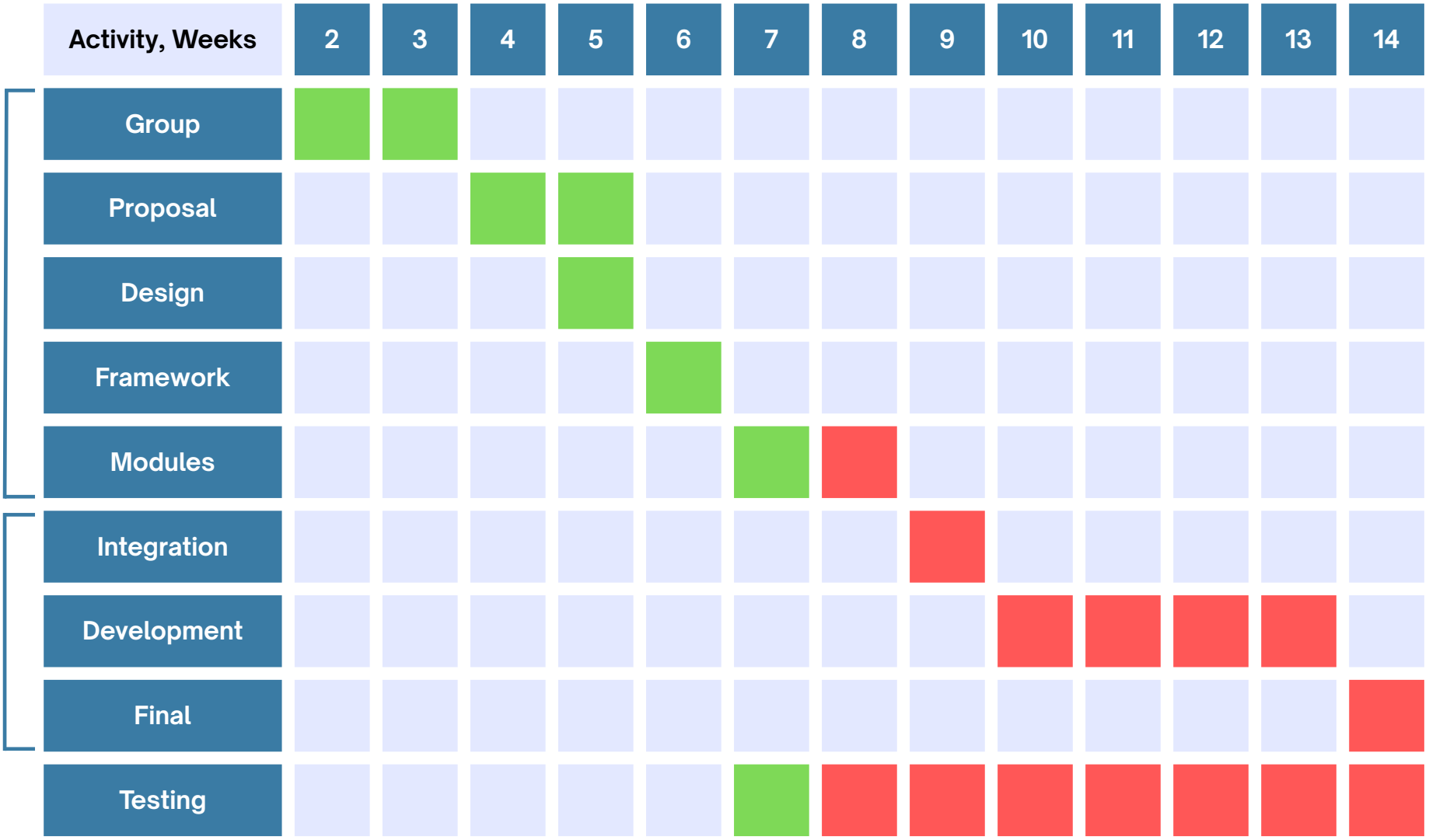
Testing going on with the implemented Modules and Documentation will given in the later presentation.

Current Status :



Work Done

Work Remains



Revised Timeline :

Activity, Weeks	2	3	4	5	6	7	8	9	10	11	12	13	14
Group													
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Development													
Final													
Testing													

Edited Time Periods

Thank You

Welcome to
Demonstration
session